

Bruce Cheng

AI Application Engineer / AI Solution Builder

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SUMMARY

AI Application Engineer building practical GenAI workflows and internal products across semiconductor, legal, engineering, and manufacturing domains. Experienced in on-premise LLM serving, RAG/agent workflows, OCR/ASR, document intelligence, and manufacturing AI.

EXPERIENCE

- Information Engineer (Generative AI / LLM Projects)** Aug. 2025 – Present
PixArt Imaging Inc. Hsinchu, Taiwan
- Build on-premise GenAI workflows for Legal, R&D, engineering, and internal knowledge scenarios under strict semiconductor data-security constraints.
 - Designed an internal GenAI workspace around vLLM model serving, document/image utilities, and task-oriented AI flows for engineering teams.
 - Built offline legal comparison, patent translation, meeting-intelligence, and ASC-to-C workflows using OCR/ASR, local LLMs, validation checks, Python tools, and review/download support.
- AI Engineer / Project Leader** Aug. 2021 – Aug. 2025
AUO Corporation - Advanced Manufacturing Center Taichung, Taiwan
- Led cross-functional AI projects connecting models, data pipelines, explainability, and internal tools to manufacturing decision workflows.
 - Improved cross-factory yield by 1.43% and reduced defects by 12.62% through Golden Path optimization and LIME-based AI parameter recommendation.
 - Evolved an intelligent assistant platform from RAG into a tool-using multi-agent workflow with SQL/API access and process-knowledge reasoning.
 - Developed a preventive maintenance workflow with LSTM and a custom trend-consistency objective, reducing defect prediction RMSE by 20%.
- Research Intern** Feb. 2019 – Jul. 2019
Academia Sinica - GIS Center Taipei, Taiwan
- Developed a web-based address normalization workflow for heterogeneous Taiwan address datasets; awarded Best Student Paper at TGIS Conference.

PROJECTS

- ASC Merge & C Code Generator** | *Open WebUI, Python, Native Tool Calling, Docker, Nginx* 2026
- Converted a repetitive SD configuration workflow into a guided tool with ordered multi-file merge, encoding fallback, comment preservation, and downloadable C/ASC outputs.
- Legal Contract Comparison System** | *Python, FastAPI, vLLM, OCR, Human Review UI* 2025 – 2026
- Automated paper contract verification using OCR, LLM restructuring, multiset omission checks, and diff review to reduce manual comparison effort.
- Private AI Gym Coach** | *React, TypeScript, Cloudflare Workers, D1, LLM API* 2026
- Developed a private workout-tracking product with selective AI support for exercise mapping and coaching prompts based on historical training volume.

TECHNICAL SKILLS

Generative AI: vLLM, On-Premise LLM, RAG, LangGraph Agents, Tool Calling, Neo4j, Prompt Engineering, OCR/ASR
Backend/Data: Python, FastAPI, Django, REST APIs, Docker, Cloudflare Workers/D1, AWS Bedrock, SQL, Databricks, ETL
ML/Product Tools: LSTM, Custom Loss Functions, LIME, Genetic Algorithms, React, TypeScript, Streamlit, Git, Linux

EDUCATION

- National Yunlin University of Science and Technology** Yunlin, Taiwan
Master of Science in Industrial Engineering and Management 2019 – 2021
National Yunlin University of Science and Technology Yunlin, Taiwan
Bachelor of Science in Industrial Engineering and Management 2015 – 2019